

DATA ANALYST CASE STUDY — Restaurant Market & Performance Analytics

Company	Zomato	Industry	Zomato— Restaurant Discovery & Delivery
Problem	Leadership can't tell which markets, price segments & features drive ratings — decisions run on guesswork.	Tools	Python SQL Power BI
Difficulty	Intermediate	Est. Time	8–10 hours

1. Business Background

Zomato Technologies is a mid-sized restaurant discovery and food-delivery platform (8 years old) letting users search restaurants, book tables, and order delivery in one app. Started in India, now live in 15 countries. Its core asset is a 9,551-restaurant listings database that powers the recommendation engine and premium-placement revenue.

Growth is strong in India (90% of listings) but inconsistent elsewhere; some newer markets show low delivery/booking adoption. Leadership currently has no shared, data-backed view of what drives ratings and where to invest next — decisions are made anecdotally in meetings.

2. The Business Problem

The VP of Strategy needs answers on: (1) which cities are under-penetrated relative to restaurant density, (2) whether delivery/booking features actually correlate with higher ratings, (3) which price segments/cuisines are most reliable, and (4) how large the "never rated" problem is. Without this, expansion and partner-incentive budgets are being planned on intuition.

Stakeholders: VP of Strategy & Expansion (market prioritization) · Head of Partnerships (delivery/booking incentives) · PM, Recommendations (rating signals) · You, Data & Analytics.

3. Your Role

You are hired as a Data Analyst at Zomato, reporting to the Head of Analytics. Own the pipeline end-to-end: clean the raw export, answer stakeholder questions in SQL, and deliver a reusable Power BI dashboard. You're accountable for data quality (documented decisions), analytical accuracy (every dashboard number traceable to a query), business framing (write for a VP, not a data scientist), and a working — not static — dashboard.

4. The Dataset

Single table zomato.csv — one row per restaurant listing, 9,551 rows × 21 columns, pulled from the platform's production DB.

Column	Type	Meaning
Restaurant ID / Name	Int / Text	Unique ID and name of the restaurant
Country Code	Integer	Numeric country code — no lookup table provided (15 unique codes)
City / Locality / Address	Text	Location fields at increasing granularity
Longitude / Latitude	Float	Geo-coordinates
Cuisines	Text (CSV)	Cuisines served; 9 rows blank
Average Cost for two	Integer	Approx. cost for two, in local currency
Currency	Text	Currency label — mismatched for several countries (data defect)
Has Table booking / Has Online delivery / Is delivering now	Yes/No	Service-availability flags
Price range	Int (1–4)	1=Low ... 4=Premium
Aggregate rating	Float (0–5)	0 = not yet rated
Rating color / Rating text	Text	UI label tied to the rating band, e.g. "Excellent"
Votes	Integer	Number of ratings submitted

Data Quality Issue	Action Required
Cuisines has 9 missing values	Fill with "Unknown" or drop — justify the choice
No Country Code → Name lookup provided	Build the mapping yourself (15 codes)
Currency label mismatched to actual country	Flag as source-system defect; standardize or footnote
Rating = 0 / Votes = 0 mixed with real ratings	Exclude unrated rows before computing averages

Data Quality Issue	Action Required
Same restaurant brand repeated across cities	Decide: analyze per-location or roll up to brand
Inconsistent casing/whitespace in text fields	Standardize before grouping

5. Task Breakdown — 3 Phases

Phase 1 — Python (Cleaning & EDA)

Load & inspect nulls/dtypes → build Country Code→Name mapping → handle missing Cuisines → convert Yes/No flags to booleans → create price_range_label & rating_bucket → exclude unrated (0/0) rows before averaging → produce 4+ labeled charts (country counts, rating distribution, rating by price range, top cuisines) → write 3+ business insights.

Phase 2 — SQL (Business Queries)

Basic: restaurants per country · avg rating/votes per city (20+ restaurants) · % never-rated per country. Intermediate: delivery/booking vs. rating · price range vs. rating & votes · top 10 cuisines by count & rating · best-rated cuisines (min. 50 restaurants). Advanced: top 3 cities per country by rating · top 10 most-voted restaurants · rating-health distribution (% by bucket) · India city-level breakdown.

Phase 3 — Power BI (Dashboard)

Visuals: country footprint map/bar · KPI cards (Total Restaurants, Avg Rating, % Unrated, % Delivery, % Booking, Avg Cost) · rating-bucket chart · price vs. rating · city drill-down table · top cuisines/restaurants. Data model: star schema (Country, City, Cuisine dimensions), not one flat table. Dashboard must let a VP see, unaided: best/largest markets, delivery-vs-rating link, and where the unrated problem is worst.

6. Business Questions to Answer

1. Which 3 cities should Zomato prioritize for its next partner-acquisition push, and why?
2. Is there a measurable ratings benefit to delivery/table booking — enough to justify subsidizing onboarding?
3. Which price segment gives the best combination of rating and popularity (votes)?
4. Which cuisines should the recommendation engine favor, by volume + rating reliability?
5. How big is the "never rated" problem, and does it vary meaningfully by country?
6. What do the top 10 most-voted restaurants have in common (city, cuisine, price)?
7. Is overall quality improving, or is this a large catalog of mediocre restaurants?

7. Deliverables & Submission Format

- .ipynb — Python notebook (Phase 1: cleaning + EDA + charts + written insights)
- .sql — All Phase 2 queries, commented with the business question each answers
- .pbix — Power BI file: relational data model, DAX measures, Phase 3 dashboard pages
- 1-page insight summary (PDF/Word) for the VP: key findings + 3 concrete recommendations

8. Evaluation Rubric

Phase	Wt.	Good	Average
Python — Clean & EDA	30%	Nulls/dupes handled w/ justification; labeled charts; 3+ insights	Undocumented cleaning; unlabeled charts
SQL — Queries	30%	Correct joins/aggregation; business-language answers	SELECT * misuse; no interpretation
Power BI — Dashboard	25%	Star schema + DAX measures; answers questions at a glance	Flat table; hardcoded numbers
Insight Summary	15%	Concise, correct numbers, clear recommendations	Restates charts, no recommendations

9. Hints / Guided Notes

- Check `df.info()` and `df.isnull().sum()` before cleaning anything.
- Country-code mapping is just a 15-entry Python dict.
- Always filter out 0-vote restaurants before averaging ratings.
- Use `HAVING COUNT(*) >= N` in SQL to avoid tiny-sample averages.

- Build Country/Cuisine dimension tables in Power BI — don't import one flat table.
- Sanity-check every dashboard number against a SQL query before presenting.